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PYROTECHNIC MISFIRE DETECTION TECHNIQUES

Abstract

A pyrotechnic misfire detection system includes one or more sensors that may acquire sensor data indicative of a pyrotechnic ignition event at one or more launch devices and a controller. The controller may receive the sensor data from the one or more sensors, determine whether the sensor data corresponds to pyrotechnic launch instructions based on the sensor data, and determine that the one or more launch devices has misfired in response to the sensor data not corresponding to the pyrotechnic launch instructions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to and the benefit of U.S. Provisional Application No. 63/559,423, entitled “PYROTECHNIC MISFIRE DETECTION TECHNIQUES,” filed on Feb. 29, 2024, which is incorporated by reference herein in its entirety for all purposes.

BACKGROUND

[0002] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be noted that these statements are to be read in this light and not as admissions of prior art.

[0003] Amusement parks or theme parks may include various entertainment attractions useful in providing enjoyment to guests of the amusement parks. For example, the attractions may include a ride attraction (e.g., closed-loop track, dark ride, thriller ride, or other similar ride), and the attraction may be part of a themed environment that may be traditionally established using equipment, furniture, building layouts, props, decorations, displayed media, and so forth. These themed environments may also incorporate pyrotechnics for aesthetic and/or entertainment purposes.

BRIEF DESCRIPTION

[0004] Certain embodiments commensurate in scope with the originally claimed subject matter are summarized below. These embodiments are not intended to limit the scope of the disclosure, but rather these embodiments are intended only to provide a brief summary of certain disclosed embodiments. Indeed, the present disclosure may encompass a variety of forms that may be similar to or different from the embodiments set forth below.

[0005] In an embodiment, a pyrotechnic misfire detection system includes one or more sensors that may acquire sensor data indicative of a pyrotechnic ignition event at one or more launch devices and a controller. The controller may receive the sensor data from the one or more sensors, determine whether the sensor data corresponds to pyrotechnic launch instructions based on the sensor data, and determine that the one or more launch devices has misfired in response to the sensor data not corresponding to the pyrotechnic launch instructions.

[0006] In an embodiment, a pyrotechnic misfire detection method includes receiving pyrotechnic launch instructions. The method also includes initiating a pyrotechnic display based on the pyrotechnic launch instructions. Further, the method includes receiving sensor data indicative of a pyrotechnic misfire and generating a notification of the misfire.

[0007] In an embodiment, a pyrotechnic misfire detection system includes one or more launch devices, the one or more launch devices including at least one launch tube and ignition circuitry that may launch a pyrotechnic from the at least one launch tube based on pyrotechnic launch instructions. The pyrotechnic misfire detection system may also include one or more sensors that may acquire sensor data at the one or more launch devices. Additionally, the pyrotechnic misfire detection system may include a controller that may transmit the pyrotechnic launch instructions to the one or more launch devices, receive the sensor data from the one or more sensors, and determine that the one or more launch devices has misfired based on the sensor data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] These and other features, aspects, and advantages of the present invention will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings,

wherein:

[0009] FIG. 1 is a schematic diagram of an embodiment of a pyrotechnic misfire detection system, in accordance with present techniques;

[0010] FIG. 2 is a schematic diagram of an embodiment of a pyrotechnic misfire detection system showing sensor data from launch devices of a pyrotechnic display, in accordance with present techniques;

[0011] FIG. 3 is a block diagram of a pyrotechnic misfire detection system including a pyrotechnic controller and a launch device, in accordance with present techniques;

[0012] FIG. 4 is an illustration of a pyrotechnic controller including a display for presenting status information of launch devices of a pyrotechnic display, in accordance with present techniques;

[0013] FIG. 5 is a flow diagram of a real-time pyrotechnic misfire detection method, in accordance with present techniques;

[0014] FIG. 6 is a flow diagram of a method for performing mitigation tasks in response to detecting a misfire of a launch device of a pyrotechnic display, in accordance with present techniques; and

[0015] FIG. 7 is a flow diagram of a method for detecting misfires of launch devices of a pyrotechnic display, in accordance with present techniques.

DETAILED DESCRIPTION

[0016] One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

[0017] When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

[0018] Pyrotechnic displays may include multiple (e.g., thousands) of launch devices that launch fireworks according to a particular ignition sequence to accomplish a desired aesthetic result. In some cases, individual launch devices or a grouping of launch devices may be individually addressed to permit more complex ignition sequences. For example, certain fireworks may be designed to be launched in conjunction with one another to create particular shapes, colors, or other pyrotechnic effects. Thus, firework launching instructions may include instructions for launching particular fireworks at particular times. However, coordinating a large pyrotechnic display may be complex. The pyrotechnic display may, for instance, supplement or be synchronized with audio performances, countdowns, or other elements. Coordinators of a pyrotechnic display may need to select appropriate launch devices and projectiles, location of the launch devices, timings of each pyrotechnic ignition, and so on to accomplish the desired aesthetic. Further, launching conditions may be constrained by safety concerns.

[0019] In some cases, individual launch devices of a pyrotechnic display may misfire due to ignition faults, controller errors, firework quality, or other malfunctions. Undetected misfires may lead to further technical faults that may compromise aesthetic qualities in subsequent displays. In some cases, misfires may be detected manually via inspection of each launch device by technicians before or after a new pyrotechnic display. However, manual inspection of each launch device may lead to delays, added complexity, or other costs, especially for large pyrotechnic displays with numerous launch devices.

[0020] Accordingly, as will be described in more detail below, the present disclosure provides techniques to efficiently detect misfires of launch devices of a pyrotechnic display. A misfire detection system for a pyrotechnic display may detect misfires based on sensor data indicative of launch device firing or misfiring and correlation with pyrotechnic instructions sent to the launch devices. In particular, the pyrotechnic misfire detection system may compare sensor data of each launch device to corresponding pyrotechnic instructions for the launch device to determine whether the sensor data is reflective of (e.g., aligning with or not aligning with) the corresponding pyrotechnic instructions. By way of example, the sensor data may include vibration data acquired by a vibration sensor at a particular launch device, and the vibration data may be time-aligned and compared to launch instructions (e.g., launch instructions concurrent with the vibration data) for the launch device to detect a successful launch or, alternatively, a misfire of the launch device.

[0021] Additionally or alternatively, the sensor data may be compared to test data to determine statuses of the launch devices. For example, the misfire detection system may operate under a test mode in which sensor data is acquired of launches that were verified as successful (e.g., by a technician). As such, the sensor data of the successful launches may form a baseline, and the pyrotechnic controller may compare subsequently acquired sensor data to the baseline to determine misfires and/or successful launches. Sensor data of a misfire may deviate from reference sensor data of a successful launch, for instance.

[0022] Upon detection of a successful launch from a launch device, an indication of the successful launch may be displayed by the misfire detection system. Alternatively, if a misfire is detected, a corresponding indication of the misfire may be displayed. As such, launch statuses of each launch device may be determined by the system in real-time and/or remotely, obviating a need for manual inspection of each launch device. Additionally, in some embodiments, the pyrotechnic controller may instruct a substitute (e.g., back-up) launch device to fire in response to detection of a misfire, such that an intended aesthetic of the pyrotechnic may be maintained. Thus, challenges associated with misfires of one or more launch devices may be mitigated. Additionally, it should be noted that while pyrotechnic displays, pyrotechnic misfires, pyrotechnic ignition events are described herein, the disclosed techniques may be applied to other launch mechanisms, such as compressed air or gas launch systems. For example, the sensor data may be indicative of an electromechanical failure (e.g., valve failure) of a compressed gas launch system.

[0023] With the foregoing in mind, FIG. 1 illustrates a system **10** including a pyrotechnic controller **12** and an array **14** of launch devices **16**. The launch devices **16** may include one or more types of launch devices suitable for launching pyrotechnics during a pyrotechnic display. Pyrotechnics may include fireworks, mortars, rockets, roman candles, fountains, firecrackers, and the like. In some cases, one or more of the launch devices **16** may be loaded with a projectile pyrotechnic, and each pyrotechnic may have a shape (e.g., artillery shell, mine), a chemical composition, or other qualities that contribute visual elements to a pyrotechnic show. For example, based on launch timing of the launch devices **16** and pyrotechnics loaded in the launch devices **16**, a pyrotechnic display may include bright twinkles (e.g., a glitter), a light trail from the launch device to a point in the sky, a palm tree, a spherical flower shape, or other visual effects. The launch devices **16** and associated pyrotechnics may also be chosen to produce audio elements, such as loud bangs (e.g., “reports” or “salutes”), or a combination of audio and visual elements. Further, the launch devices **16** may include one or more electrical matches or other ignition starters that cause the launch devices **16** to fire loaded pyrotechnics. Each electrical match may include, for example, combustible material that ignites when a current flows through the electrical match. In the illustrated embodiment, the launch devices **16** are grouped such that they form the array **14** of launch devices **16**. Each launch device **16** of the array **14** may, for example, be positioned on or associated with a base **17**, such as a structural frame (e.g., mortar rack), or a sand-filled structural support (e.g., mortar trough) to support the launch devices **16** in an upright or angled position. As illustrated, an individual launch device **16** may include one or more launch tubes **19**. The launch

tube **19** may form a passage into which an individual pyrotechnic **21** can be loaded and subsequently launched from.

[0024] The pyrotechnic controller **12** may provide pyrotechnic launch instructions **18** to the array **14** of launch devices **16**. The pyrotechnic launch instructions **18** may include a current or other electrical characteristic that may cause one or more launch devices **16** of the array **14** to fire pyrotechnics **21** via, for example, electrical matches of the launch devices **16**. The pyrotechnic controller **12** may send the pyrotechnic launch instructions **18** to the array **14** to establish a launch schedule or launch timing, such that each launch device of the array **14** fires a pyrotechnic **21** at a certain time point or time range during a pyrotechnic display. An individual launch device **16** may be independently addressed to launch different pyrotechnics at different times and/or in series or in parallel with one another. In some cases, the launch instructions **18** and pyrotechnic loading may be set such that adjacent launch tubes **19** do not launch pyrotechnics simultaneously to avoid interference. In one example, pyrotechnic launch instructions **18** may cause the launch devices **16** to fire pyrotechnics **21** at the chorus of an accompanying music performance, at a finale of a pyrotechnic display, and so on. For example, in response to the pyrotechnic launch instructions **18**, the launch devices **16** of the array **14** may ignite in rapid succession to produce a barrage of aerial pyrotechnics.

[0025] The array **14** may also include one or more sensors **20** that may acquire pyrotechnic sensor data **22** of the launch devices **16**. The one or more sensors **20** may include microphones, pressure sensors, temperature sensors, vibration sensors, accelerometers, or other sensors suitable for acquiring sensor data useful in determining successful launches or misfires of the launch devices **16**. In the illustrated embodiment, the one or more sensors **20** include sensors coupled to each of the launch devices **16**. The launch devices **16** may include launch tubes **19**, for example, and a sensor **20** may be placed on the exterior of each launch tube **19**. As mentioned, in some embodiments, each launch device **16** of the array **14** may be supported by the base **17**. In these embodiments, the one or more sensors **20** may include a sensor **20** arranged on a supporting structure or the base **17** of the array **14** and, as such, the sensor **20** may acquire pyrotechnic sensor data **22** of multiple launch devices **16** supported by the base **17**. In another example, the one or more sensors **20** may include an optical sensor positioned to capture infrared sensor data of each launch device **16**. In another example, the one or more sensors **20** may include a camera positioned to capture image data (e.g., video) of each launch device **16**. In any case, the pyrotechnic sensor data **22** acquired by the one or more sensors **20** may be sent to the pyrotechnic controller **12** for analysis.

[0026] While FIG. 1 illustrates the system **10** as including the array **14** of launch devices **16**, in some embodiments, the system **10** may include multiple arrays **14** with numerous launch devices **16**. FIG. 2 illustrates the system **10** in which the pyrotechnic controller **12** receives pyrotechnic sensor data from a first array **30** (e.g., array of launch tubes **19**), a second array **32**, a third array **34**, and a fourth array **36** during a pyrotechnic display. The illustrated arrangement is by way of example, and it should be understood that additional or fewer launch devices **16** or arrays **14** may be used. FIG. 2 illustrates the system **10** at a particular time interval, such as a one-second snapshot, during a pyrotechnic display. Further, while the illustrated arrangement shows an individual sensor **20** associated with each array **14**, the sensor **20** may be coupled to each individual launch device **16**, or alternate launch devices **16**, in an embodiment.

[0027] In the illustrated embodiment, each of the second array **32** and the third array **34** successfully launch a corrected pyrotechnic and, as such, second pyrotechnic sensor data **38** of the second array **32** and third pyrotechnic sensor data **40** of the third array **34** each include sensor data indicative of a successful launch. For example, the second pyrotechnic sensor data **38** indicative of a successful launch may include pressure data indicative of a rapid change in pressure (e.g., within a pyrotechnic or gas-compressed launch tube), vibration data indicative of a rapid vibration of the second array **32**, infrared signatures indicative of an ignition flash in one or more tubes of the

second array **32**, other sensor data indicative of a successful launch, or a combination thereof. In certain cases, the sensor data indicative of a successful launch may include data that is at or above a set threshold. In an example, a vibration amplitude from a vibration sensor associated with an individual array **14** and/or an individual launch device **16** being at or above a threshold associated with a successful launch may be indicative of a successful fire event. In another example, a light intensity from an optical sensor associated with an individual array **14** and/or an individual launch device **16** may be at or above a threshold associated with a successful launch. The optical sensor may be set to capture or select optical data from a tube opening or just outside of the tube opening to track launches or misfires.

[0028] On the other hand, the first array **30** and the fourth array **36** do not successfully launch within the illustrated time interval. As a result, first pyrotechnic sensor data **42** of the first array **30** and fourth pyrotechnic sensor data **44** of the fourth array **36** each include sensor data not indicative of a launch. In certain cases, the sensor data indicative of no launch may include data that is below a set threshold. In an example, a vibration amplitude from a vibration sensor associated with an individual array **14** and/or an individual launch device **16** being below a threshold associated with a successful launch may be indicative of a misfire. In another example, a light intensity from an optical sensor associated with an individual array **14** and/or an individual launch device **16** may be below a threshold associated with a successful launch.

[0029] As may be appreciated, the first pyrotechnic sensor data **42** and the fourth pyrotechnic sensor data **44** may each include pressure data not indicative of a rapid change in pressure, vibration data not indicative of a rapid vibration, and so on. Alternatively, the pressure data may be indicative of a rapid change in pressure, but the rapid change in pressure may not be consistent with a successful launch, may include a rapid vibration inconsistent with a successful launch, and so on. In some cases, the pyrotechnic controller **12** may, based on the pyrotechnic sensor data from an array **14** (e.g., during a time interval) and pyrotechnic instructions sent to the array (e.g., for the time interval), determine that a misfire has occurred in the array **14** (e.g., may determine a misfire launch status for the array **14** or one or more tubes **19** of the array **14**). It should be noted that in other cases, however, the sensor data not indicative of a launch may not indicate misfires (e.g., of the first array **30** and/or the fourth array **36**), as the pyrotechnic instructions (e.g., sent to the first array **30** and the fourth array **36**) may not have included instructions to launch during the illustrated time interval. Thus, as generally discussed herein, the misfire detection may include a determination of whether a launch status of the launch device **16** aligns with the pyrotechnic launch instructions for one or more time points.

[0030] FIG. **3** illustrates a block diagram of the pyrotechnic controller **12** and an example launch device **16** of the preceding figures. The launch device **16** may include ignition circuitry **52**, such as the electrical matches, that facilitate launching pyrotechnics in response to received electrical signals. The launch device **16** may also include one or more sensors **20** that are launch device-coupled sensors **54**, such as microphones, pressure sensors, temperature sensors, vibration sensors, accelerometers, or other sensors suitable for acquiring sensor data useful in determining successful launches or misfires of the launch device **16**, as described herein. Additionally, the launch device **16** may include an ignition indicator **56**, and the ignition indicator may display an indication of a misfire or successful launch in response to a misfire or successful launch of the launch device **16** being detected. For example, the one or more sensors **54** may generate sensor data, the pyrotechnic controller **12** may determine that the sensor data is indicative of a successful launch, and the pyrotechnic controller **12** may send a signal to the ignition indicator **56** of the launch device **16** indicative of instructions to display an indication of the successful launch. The ignition indicator **56** may include a light-emitting diode (LED), a display, or the like, and may, for example, display a green indication indicative of a successful launch or, alternatively, a red indication indicative of a misfire. This may allow a technician to efficiently ascertain a status of the launch device **16** without closely inspecting the launch device **16**.

[0031] The pyrotechnic controller **12** may include, for example, processing circuitry **58**, a memory **60**, a firing unit **62**, a display **64**, and communication circuitry **66**. The processing circuitry **58** may include one or more suitable processors that can execute instructions for carrying out the presently disclosed techniques, such as a general-purpose processor, system-on-chip (SoC) device, an application-specific integrated circuit (ASIC), a processor of a programmable logic controller (PLC), a processor of an industrial PC (IPC), or some other similar processor configuration. These instructions are encoded in programs or processor-executable code stored in a tangible, non-transitory, computer-readable medium. The memory **60** may include one or more storage devices, and may store machine-readable and/or processor-executable instructions (e.g., firmware or software) for the processing circuitry **58** to execute, such as instructions relating to generating pyrotechnic instructions, instructions relating to determining launch statuses, or instructions for generating back-up launch instructions. In some cases, the instructions may be generated using one or more machine learning (ML) models, computer vision functions, and the like to determine a launch status based on sensor data and launch instructions. For example, the instructions may relate to using computer vision to determine a launch status based on sensor data from one or more cameras. As such, the memory **60** may store, for example, control software, look up tables, configuration data, ML models, and so forth. The memory **60** may include a tangible, non-transitory, machine-readable-medium, such as a volatile memory (e.g., a random access memory (RAM)) and/or a nonvolatile memory (e.g., a read-only memory (ROM), flash memory, hard drive, and/or any other suitable optical, magnetic, or solid-state storage medium). The memory **60** may store pyrotechnic display configurations, launch schedules, launch statuses, and so on. The memory **60** may also store instructions to determine a launch status based on sensor data and launch instructions, and the processing circuitry **82** may execute the instructions.

[0032] The firing unit **62** may include a current source, power source, or the like that facilitates the generation and/or transmission of electrical signals as launch instructions for the launch device **16**. For example, the firing unit **62** may, when instructed by the processing circuitry **58**, generate and send one or more currents or other electrical qualities to the launch device **16** that cause the ignition circuitry **52** to ignite and actuate the launch device **16**. While the illustrated example shows the firing unit **62** as part of the controller **12**, the firing unit **62** may, in embodiments, be coupled to the launch device **16** and/or the array **14**. In some cases, the firing unit **62** may allow manual control by a technician via a panel, electrical box, or other interface. For example, a technician may instruct the firing unit to generate and send instructions to cause a corresponding back-up launch device to actuate in response to a misfire in the launch device **16**. Additionally or alternatively, the pyrotechnic controller **12** may include communication circuitry **66**, and the communication circuitry **66** may facilitate communication of launch instructions, sensor data, display instructions, and the like to the launch device **16** or other devices via wireless communication with corresponding communication circuitry **67** of the launch device **16**.

[0033] The pyrotechnic controller **12** may also be communicatively coupled to one or more additional sensors **68**, such as cameras or optical sensors that are positioned to acquire sensor data of the launch device **16**. In an embodiment, the additional sensors **68** may generate data of the connection between the pyrotechnic controller **12** and the launch device **16**, multiple launch devices (e.g., an array **14** of launch devices), multiple arrays **14** of launch devices, or each launch device **16** of a pyrotechnic display. The additional sensors **68** may also serve as an alternative to the one or more sensors **54** of the launch device **16** in case of a technical fault or other failure of the one or more sensors **54**. In some embodiments, the additional sensors **68** may generate sensor data of the firing unit **62** such that the pyrotechnic controller **12** may determine whether a misfire is due to a fault in the firing unit **62** (e.g., the firing unit **62** failed to produce an electrical signal in response to instructions from the processing circuitry **58**).

[0034] The display **64** may display indications of a successful launch or misfire of the launch device **16**. The display **64** may include a graphical user interface (GUI), such as a touch-screen

GUI, and the GUI may display information regarding a present or planned pyrotechnic display, such as launch device locations, launch device types, launch instruction schedules, and so on. In particular, the GUI displayed by the display **64** may include indications of a launch status of the launch device **16** as determined by the pyrotechnic controller **12**.

[0035] The display **64** may, in some embodiments, display launch status indications for multiple launch devices. FIG. **4** illustrates such an embodiment of system **10**. In the illustrated embodiment, the display **64** of the pyrotechnic controller **12** displays status indicators of a first array, a second array, a third array, a fourth array, a fifth array, and a sixth array of a pyrotechnic display **70**. In the illustrated embodiments, each array of the pyrotechnic display **70** includes a sensor that generates respective sensor data of the array, and the respective sensor data is sent to the pyrotechnic controller **12**, as described herein. The pyrotechnic controller **12** may then, for each respective array, determine whether a successful launch or misfire has occurred for the respective array based on a determination of whether the respective sensor data corresponds to launch instructions for the respective array. By way of example, sensor data of the second array may be indicative of rapid vibration at a time interval 20 seconds into a pyrotechnic display. The pyrotechnic controller **12** may determine that the sensor data aligns with launch instructions for the second array (e.g., instructions to fire at T-20 seconds), and may thus determine a successful launch for the second array. This determination may be displayed as part of a second array indicator **72**, illustrated here as a check mark following a listing of the second array.

[0036] Alternatively, the pyrotechnic controller **12** may determine, based on sensor data and launch instructions, that a misfire has occurred in one or more launch devices. Sensor data of the fourth array, for example, may be indicative of a lack of rapid vibration for a certain time interval or for the duration of the pyrotechnic display. The pyrotechnic controller **12** may determine that the sensor data of the fourth array does not align with launch instructions for the fourth array (e.g., launch at T-25 seconds), and may thus determine a misfire for the fourth array. In another example, the sensor data of the fourth array may be indicative of rapid vibration at a time interval not aligning with the time interval for launch included in the launch instructions (e.g., rapid vibration at T-5 seconds, instructions to launch at T-20 seconds), and the pyrotechnic controller **12** may accordingly determine a misfire of the fourth array. In any case, the determination of the misfire of the fourth array may be displayed as a fourth array indicator **74**, here illustrated as an “X” following a listing of the fourth array.

[0037] In some embodiments, the pyrotechnic controller **12** may not determine that a misfire or successful launch has occurred in one or more launch devices. For example, sensor data from a launch device may be only partially indicative of a successful launch, or may not be indicative of a threshold overlap or alignment with the launch instructions. In such cases, the pyrotechnic controller **12** may instead determine a confidence interval associated with the launch device based on sensor data and launch instructions for the launch device. Such a determination may arise as a result of a faulty but nonetheless evident launch, sensor data for a launch device being influenced by ignitions of neighboring launch devices, and so on. By way of example, sensor data of the sixth array may be indicative of some vibration (e.g., less than a threshold launch vibration) at a time interval nearly aligning with the launch instructions for the sixth array (e.g., instructions to launch at T-20 seconds, some vibration at T-21 seconds). Accordingly, the determination of a confidence interval (e.g., neither a misfire nor successful launch) may be displayed as a sixth array indicator **76**, here illustrated as a question mark following a listing of the sixth array. In some embodiments, a numerical confidence interval (e.g., 50%, 75%) may be determined by the pyrotechnic controller **12** based on launch instructions, sensor data, threshold time or sensor values input via a graphical user interface, and so on, and may be indicated via the display **64** accordingly. In any case, the displayed indication may indicate, to a technician, that the launch device may need to be manually inspected for a misfire at the conclusion of the pyrotechnic display, for example.

[0038] In certain embodiments, the determination of misfires, successful launches, and/or

confidence intervals, and corresponding indications of the determinations, may update dynamically throughout a duration of a pyrotechnic display. This may be advantageous for pyrotechnic displays that include launch devices or arrays that may fire at multiple time intervals throughout the pyrotechnic display, as a launch device may successfully fire and subsequently misfire, may misfire and later successfully fire, and so on. For example, the pyrotechnic controller **12** may determine a successful launch of the second array at a first time interval (e.g., T-10 seconds), but may determine a misfire of the second array at a second time interval (e.g., T-20 seconds). As such, the second array indicator **72** may display a check mark after the first successful launch, and may update to display an “X” after the determined misfire. Additionally or alternatively, the display **64** may display a matrix, table, or similar mapping of launch devices to instructed launches, and may display an indication of whether each instructed launch was determined as a successful launch or a misfire. The display **64** may display separate column entries for each instructed launch, for instance.

[0039] In some cases, certain components and functions may be included and performed by a device or system separate from the pyrotechnic controller **12**. For example, the pyrotechnic controller **12** may include the firing unit **62**, the communication circuitry **66**, the processor **58**, and/or the memory **60** for communicating launch instructions, but may not receive sensor data, determine a launch status, instruct backup tubes to launch, and the like. Instead, such functions may be performed by a standalone misfire detection system with one or more of the processor **58**, memory **60**, firing unit **62**, display **64**, and communication circuitry **66**. In such examples, the pyrotechnic controller **12** may use the launch instructions to launch pyrotechnics, while the standalone misfire detection system may use the launch instructions to determine a misfire, for instance.

[0040] With the foregoing in mind, FIG. **5** is a flowchart of a method **100** for detecting misfires in a pyrotechnic display and is discussed with reference to FIGS. **1-4**. While described as being performed by the pyrotechnic controller **12**, one or more of the steps of the method **100** may be performed by a standalone misfire detection system. The process begins with initiating the pyrotechnic display (block **102**). Initiating the pyrotechnic display may include, for example, the loading of launch instructions for each launch device and/or array of launch devices into a memory of the pyrotechnic controller (e.g., the memory **60** of the pyrotechnic controller **12**), loading projectiles or other pyrotechnics into each launch device, initializing an audio output synchronized with the pyrotechnic display, starting timers used for determining a correspondence between sensor data and launch instructions, and so on. Initiating the pyrotechnic display may also include sending first launch instructions (e.g., first currents) to one or more launch devices to begin the pyrotechnic display.

[0041] After the pyrotechnic display is initiated (block **102**), sensor data from one or more sensors **20** may be received at the pyrotechnic controller **12** (block **104**). As described herein, the sensor data may include audio values, pressure values, temperatures, vibration data, acceleration data, and the like of one or more launch devices or arrays of launch devices. The received sensor data may be aligned with corresponding launch instructions (block **106**). Aligning the received sensor data with corresponding launch instructions may include mapping of the received sensor data to launch instructions. For example, sensor data received from a sensor attached to the first launch device may be aligned with launch instructions for the first launch device stored in the memory **60** of the pyrotechnic controller **12**, sensor data received from a sensor attached to a mortar rack of a first array of launch devices may be aligned with launch instructions for the first array stored in the memory **60** of the pyrotechnic controller **12**, and so on. Additionally, aligning the received sensor data with corresponding launch instructions may include a time alignment. For example, the pyrotechnic controller **12** may assign a timestamp (e.g., based on a timer started during initiation) to received sensor data, and the received sensor data may be compared to launch instructions at the timestamp. Further, a delay may be determined during a testing procedure of the pyrotechnic

display characterizing a time delay between the sending of launch instructions and the reception of sensor data characterizing a successful launch. This delay may be subtracted from the timestamp to account for delays in the transmission of instructions to the launch devices and/or delays in receiving sensor data from sensors of the launch devices.

[0042] In some embodiments, the sensor data may be received in real-time during a pyrotechnic display, and may be compared to a threshold to determine a successful launch or misfire for each launch device. For example, during a testing procedure of the pyrotechnic display, a threshold of vibration (e.g., 100 hertz) may be established as a threshold minimum indicative of a launch. As such, if sensor data indicative of vibration above the threshold is received and mapped to a time interval of the launch instructions that includes instructions to launch, the pyrotechnic controller **12** determines a successful launch (block **110**). If, however, sensor data indicative of vibration below the threshold is received and mapped to the time interval of the launch instructions that includes the instructions to launch, the pyrotechnic controller **12** may determine a misfire (block **108**). Additionally, if the received sensor data is indicative of vibration above the threshold and mapped to a time interval of the launch instructions that does not include instructions to launch, the pyrotechnic controller may determine a misfire (block **108**). In any case, the pyrotechnic controller **12** may continue to receive sensor data from the launch devices (block **104**).

[0043] FIG. **6** is a flowchart of a method **120** for performing mitigation tasks in response to determining a misfire in one or more launch devices of a pyrotechnic display, and may be performed as part of or in conjunction with the method **100** of FIG. **5**. The process may begin with determining a misfire, as generally described above (block **122**). In response to a misfire being determined for one or more launch devices, an indication of a misfire may be displayed via, for example, the display **64** of the pyrotechnic controller **12** or other suitable means (block **124**). The determination of the misfire of one or more launch devices may be displayed as an “X” following a listing of the misfired launch device or as part of a matrix of launch devices and instructed launches. In another example, a graphical user interface presented by the display **64** may include a map of the launch area (e.g., the area by which pyrotechnics are launched from) of the pyrotechnic display, and a misfire may be indicated on the map by appropriate indicators.

[0044] The pyrotechnic controller **12** may also instruct a backup launch device to fire in response to determining a misfire by, for example, augmenting the launch instructions to include instructions to fire the back-up launch device (block **126**). The backup launch device may be determined by, for example, a mapping of launch devices (e.g., primary launch devices) to corresponding back-up launch devices. The back-up launch device may be situated near one or more launch devices, and may have similar qualities (e.g., projectile type, chemical composition), such that firing the back-up launch device may preserve an intended aesthetic of the pyrotechnic display in case of a misfire. Further, the back-up launch device may include sensors to acquire sensor data of the back-up launch device, the sensor data may be sent to the pyrotechnic controller **12**, and the pyrotechnic controller **12** may determine a successful launch or misfire of the back-up launch device.

Accordingly, if a misfire is detected in the back-up launch device, the method **120** may begin anew with determining the misfire (block **122**). In addition, the back-up launch device may itself have an additional back-up device that may be instructed to launch should the back-up device misfire.

[0045] FIG. **7** is a flowchart of a method **130** for detecting misfires in a pyrotechnic display and is discussed with reference to FIGS. **1-4**. The process may begin with initiating the pyrotechnic display (block **132**). As mentioned herein, initiating the pyrotechnic display may include, for example, the loading of launch instructions for each launch device and/or array of launch devices into a memory of the pyrotechnic controller, loading projectiles or other pyrotechnics into each launch device, initializing an audio output synchronized with the pyrotechnic display, starting timers used for determining that the pyrotechnic display has concluded, and so on. Initiating the pyrotechnic display may also include sending first launch instructions (e.g., first currents) to one or more launch devices to begin the pyrotechnic display.

[0046] After the pyrotechnic display is initiated (block **132**), sensor data generated at the launch devices may be received at the pyrotechnic controller **12** (block **134**). As described herein, the sensor data may include audio values, pressure values, temperatures, vibration data, acceleration data, and the like of one or more launch devices or arrays of launch devices at a time interval. The sensor data may be used to determine a successful launch of the launch device (block **136**) of which the sensor data was acquired at the time interval via, for example, comparison of the sensor data to a threshold. The threshold may include a minimum volume, pressure minimum, vibration minimum, or the like that characterizes a successful launch. If, at the time interval, the sensor data is indicative of a successful launch, a successful launch is determined for the launch device (block **138**).

[0047] If a successful launch is not detected at the time interval (e.g., sensor data is not above the threshold for the time interval), a determination whether the pyrotechnic display has concluded is made (block **138**). The determination whether the pyrotechnic display has concluded may be made based on an expiration of a timer (e.g., a timer stored in the memory **60** of the pyrotechnic controller **12**) started during initiation of the pyrotechnic display. Alternatively, the determination whether the pyrotechnic display has concluded may be made based on technician input via, for example, a graphical user interface of the display **64** or other suitable input means. If the display has not concluded, the pyrotechnic controller **12** may continue to receive sensor data (block **134**).

[0048] If the pyrotechnic display has concluded, a misfire may be determined (block **140**). In some embodiments, this may include determining a misfire for each launch device for which a successful launch was not determined (e.g., in block **138**). In response, mitigation tasks may be performed by the pyrotechnic controller **12**. In particular, the pyrotechnic controller may provide a notification of the location of each launch device that misfired (block **142**) and/or identity of each launch device that misfired. For example, the display **64** may present a textual notification including a grid location of each launch device that was determined as having misfired, or may present a map with indications of the locations of each misfired launch device. Additionally or alternatively, the notification may include an activated indicator light on or near a launch device associated with the misfire.

[0049] While only certain features of the invention have been illustrated and described herein, many modifications and changes will occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the invention.

[0050] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for (perform)ing (a function) . . . ” or “step for (perform)ing (a function) . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A pyrotechnic misfire detection system, comprising; one or more sensors configured to acquire sensor data indicative of a pyrotechnic ignition event at one or more launch devices; and a controller configured to: receive the sensor data from the one or more sensors; determine whether the sensor data corresponds to pyrotechnic launch instructions; and determine that the one or more launch devices has misfired based on the sensor data not corresponding to the pyrotechnic launch instructions.

2. The system of claim 1, wherein the one or more launch devices are coupled to a plurality of launch tubes, each launch tube of the plurality of launch tubes comprising a pyrotechnic of a

plurality of pyrotechnics.

3. The system of claim 2, wherein determining whether the sensor data corresponds to the pyrotechnic launch instructions comprises identifying a launch status of each respective pyrotechnic.
 4. The system of claim 3, wherein determining that the one or more launch devices has misfired comprises identifying an individual launch tube of the plurality of launch tubes having a misfire launch status.
 5. The system of claim 2, wherein the one or more sensors comprise a vibration sensor coupled to each launch tube of the plurality of launch tubes.
 6. The system of claim 1, wherein the one or more sensors comprise at least one vibration sensor coupled to a launch device of the one or more launch devices.
 7. The system of claim 6, wherein the sensor data is indicative of vibration of the launch device concurrent with execution of the pyrotechnic launch instructions.
 8. The system of claim 7, wherein a first vibration amplitude above a threshold is indicative of a successful fire event and wherein a second vibration amplitude below the threshold is indicative of the misfire.
 9. The system of claim 1, comprising generating a notification indicative of the misfire.
 10. The system of claim 9, wherein the notification comprises a location or identity of a launch device, of the one or more launch devices, associated with the misfire.
 11. The system of claim 1, wherein the one or more sensors comprise a camera.
 12. A pyrotechnic misfire detection method comprising: receiving, at a pyrotechnic launch controller, pyrotechnic launch instructions; initiating, via the pyrotechnic launch controller, a pyrotechnic display based on the pyrotechnic launch instructions; receiving, via the pyrotechnic launch controller, sensor data indicative of a pyrotechnic misfire; and generating, via the pyrotechnic launch controller, a notification of the misfire.
 13. The method of claim 12, wherein the sensor data is, pressure sensor data, optical sensor data, or a combination thereof from a launch device.
 14. The method of claim 12, wherein the sensor data indicative of the pyrotechnic misfire comprises a vibration signal indicative of a misfire in one or more launch tubes of a launch device.
 15. The method of claim 12, wherein the notification comprises a location or identity of a launch device associated with the misfire.
 16. The method of claim 12, wherein the notification comprises an activated indicator light on or near a launch device associated with the misfire.
 17. A pyrotechnic misfire detection system, comprising: one or more launch devices, comprising: at least one launch tube; and ignition circuitry configured to launch a pyrotechnic from the at least one launch tube based on pyrotechnic launch instructions; one or more sensors configured to acquire sensor data at the one or more launch devices; and a controller configured to: transmit the pyrotechnic launch instructions to the one or more launch devices; receive the sensor data from the one or more sensors; and determine that the one or more launch devices has misfired based on the sensor data.
 18. The system of claim 17, wherein the controller is configured to determine that the one or more launch devices has misfired based on the sensor data being indicative of no pyrotechnic launch at a time range corresponding to an instructed pyrotechnic launch from the launch tube.
 19. The system of claim 18, wherein the sensor data comprises a deviation from a reference associated with a pyrotechnic launch.
 20. The system of claim 17, wherein the controller is configured to: augment the pyrotechnic launch instructions to include instructions to launch one or more backup launch devices in response to the determination that the one or more launch devices has misfired; and transmit the augmented pyrotechnic launch instructions to the one or more backup launch devices.
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